

Baltazar González

BIOLOGIST · BIOGEOGRAPHER · PROGRAMMER

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“Science is not a heartless pursuit of objective information; it is a creative human activity.” S. J. Gould

Current position

University of California, Merced

POSTDOCTORAL RESEARCHER

California, USA

2024 - Present

My research examines how ecological and evolutionary processes change across space and time in the context of climate change. For my post-doctoral research, I focus on pollination and seed dispersal as key processes for biodiversity maintenance and food production, using them as model systems to understand how ecological interactions respond to environmental variability at regional and global scales. In addition, I explore how evolutionary processes give rise to transition zones and how different niche metrics relate to these spatial patterns.

I combine predictive modeling with large-scale biodiversity, climate, and land-use data, integrating spatial and temporal analyses within automated, reproducible computational workflows. Looking ahead, my work aims to move beyond static projections toward interaction-based and process-informed models that capture how ecological services reorganize under climate extremes, helping identify vulnerable systems and inform climate-adaptive conservation and management.

Education

Universidad Nacional de La Plata

PH.D. IN NATURAL SCIENCES

La Plata, Argentina

2018 - 2024

- Thesis: Systematics, biogeography, and adaptation of the extant marsupial order Paucituberculata (Caenolestidae).
- Awarded highest score on thesis defense.

Universidad del Valle

B.Sc. IN BIOLOGY

Cali, Colombia

2010 - 2015

- Thesis: Demographic parameters of the Silky Shrew-opossum, *Caenolestes fuliginosus* in the Central Cordillera of Colombia.
- Awarded two grants and recognition for top 5 GPA of the career.

Peer-reviewed Publications

IN PREPARATION

Argueta-Guzmán, M.P., **González, B.**, Hutchinson, M., Gaiarsa, M.P., Rising heat waves disrupt pollination services in agriculture. *To be submitted to PNAS by the end of June 2026*

González, B., Guimaraes, P., Gaiarsa, M.P. From Tropics to Temperate: Latitudinal Vulnerability of Seed-Dispersal Networks under Climate Change. *To be submitted to Journal of Biogeography by the end of August 2026.*

SUBMITTED

González, B., Argueta-Guzmán, M.P, Mora-Sibaja, J., Salerno, V., Tang, A., Crall, J.D., McFrederick, Q., Gaiarsa, M.P. Climate-driven shifts in bee and floral vulnerability alter when and where conservation is needed. *Minor revision Communication Biology - August 2026.*

Argueta-Guzman et al. 2026. Distinct foraging goals shape floral resource use in a generalist solitary bee. *Ecology Letters In revision August 2026*

Giménez, A. L. **González, B.**, Martin, G.M. Climate culs-de-sac: range contractions and conservation mismatches in the world's southernmost bats. *Global Ecology and Biogeography In revision May 2026*

2026

Salerno, V., Tang, A.T.So., Mora, J., Argueta-Guzmán, M., **González, B.**, Peterson, S.S., McFrederick, Q.S., Gaiarsa, M.P., Crall, J. osmiaCAM: A Low-cost Imaging Monitoring System for Linking Solitary Bee Foraging, Nest Provisioning, and Environmental Sensitivity, *Integrative and Comparative Biology*, <https://doi.org/10.1093/icb/icag081>

2025

Brook, B., **González, B.**, Tomasco, I.H., Verzi, D. H., Martin, G. M. Within the forest: a new species of *Ctenomys* (Rodentia: Ctenomyidae) from northwestern Patagonia. *Journal of Mammalogy*, 106, 1, <https://doi.org/10.1093/jmammal/gyae101>

2024

Martin, G.M., **González, B.**, Brook, F. et al. Requiem for Argentine mammals: A spatial framework for mapping extinction risk. *Journal of Nature Conservation* 82. <https://doi.org/10.1016/j.jnc.2024.126759>

González, B., Ferro-Muñoz, N., Calvache, C., Rojas, D., Martin G.M. Mind the gap: New records of *Caenolestes* in the Western Andes of Colombia challenge its current biogeographic patterns. *Journal of Mammalogy*. <https://doi.org/10.1093/jmammal/gyae035>.

Lizcano, D.J., **González, B.**, Otálora-Ardila A., Ramírez-Chaves, H.E., Torres-Martínez, M.M. Mammalogy Notes cumple 10 años y celebra su contribución al conocimiento de los mamíferos de Colombia y el mundo. *Mammalogy Notes*. <https://doi.org/10.47603/mano.v10n1.422>. (Editorial)

2023

Frank, F., Volante, J., Calamari, N., Peri, P. L., **González, B.**, et al. A multi-model approach to explore sustainable food and land use pathways for Argentina. *Sustainability Science*, 18(1), 347–369. <https://doi.org/10.1007/s11625-022-01245-5>

Guerisoli, M.M., et al. Reflexiones acerca del reasilvestramiento en la Argentina. *Mastozoología Neotropical*, 30(1), 001-012. <https://doi.org/10.31687/saremMN.23.30.1.06.e0946> (Editorial)

Guerisoli, M. M. et al. Threatened conservation scientists: The aftermath of an eye-opening publication on rewilding. *Biological Conservation*, 287, 110322. <https://doi.org/10.1016/j.biocon.2023.110322> (Editorial)

2022

Martin, G.M., **González, B.**, Brook, F., Monjeau, A. Conservation biogeography of living American marsupials: Didelphimorphia, Microbiotheria, and Paucituberculata. In: Cáceres, N.C., Dickman, C.R. (eds) *American and Australasian Marsupials*. Springer, Cham. https://doi.org/10.1007/978-3-030-88800-8_25-1 (Book chapter)

González, B., Brook, F., Martin, G. M. Updated distribution and conservation perspectives of marmosine opossums from Colombia. *Hystrix, the Italian Journal of Mammalogy*. <https://doi.org/10.4404/hystrix-00489-2021>

Martin, G. M., **González, B.**, Brook, F., Cirignoli, S., Monjeau, J. A. Los vulnerables: Las áreas protegidas de argentina no protegen a los marsupiales. *Mastozoología Neotropical*, 29(1), 001-014. <https://doi.org/10.31687/saremMN.22.29.1.12.e0670>

2021

Martin, G. M., **González, B.**, Monjeau, A. Continental assessment of South American marsupial conservation priorities: A methodological approach using a spatially explicit conservation indicator. *Biological Conservation*, 256, 109045. <https://doi.org/10.1016/j.biocon.2021.109045>

Brook, F., Tomasco, I. H., **González, B.**, Martin, G. M. A New Species of *Ctenomys* (Rodentia: Ctenomyidae) from Patagonia Related to *C. sociabilis*. *Journal of Mammalian Evolution*. <https://doi.org/10.1007/s10914-021-09570-9>

2020

González, B., Soria Escobar, A. M., Rojas Díaz, V., Pustovrh, M. C., Monsalve, L. S., Rougier, G. W. The embryo of the silky shrew opossum, *Caenolestes fuliginosus* (Tomes, 1863): First description of the embryo of Paucituberculata. *Journal of Morphology*, 281(3), 326-337. <https://doi.org/10.1002/jmor.21101>

González, B., Brook, F., Martin, G. M. Variability and variation in *Rhyncholestes raphanurus* Osgood (Paucituberculata, Caenolestidae). *Revista Chilena de Historia Natural*, 93(1), 1. <https://doi.org/10.1186/s40693-020-00089-6>

Martin, G., Soibelzon, E., Ciai, D., Brook, F., **González, B.**, Negrete, J. Nuevos registros para *Lestodelphys halli* (Thomas) en la provincia de La Pampa (República Argentina). *Notas sobre Mamíferos Sudamericanos*, 01(1), 001-008. <https://doi.org/10.31687/saremNMS.20.0.16>

Ascuntar-Osnas, O., Montoya-Bustamante, S., **González, B.** Records of Streblidae (Diptera: Hippoboscoidea) in a tropical dry forest fragment in Colombia. *Biota Colombiana*, 21(1). <https://doi.org/10.21068/c2020.v21n01a02>

Niño-Reyes, A., **González, B.**, Navarro-Morales, A., Moreno, A., Velásquez-Valencia, A., Capera, G., Cepeda-Duque, J.C. New record in the diet of *Philander andersoni*. *Therya Notes*, 1(1), 14-18. https://doi.org/10.12933/therya_notes-20-5

2019

González, B., Rojas-Díaz, V., Cruz-Bernate, L. Demographic parameters of the silky shrew opossum *Caenolestes fuliginosus* (Paucituberculata, Caenolestidae) along an altitudinal gradient in the cordillera central of the Colombian Andes. *Journal of Mammalian Evolution*, 26(1), 39-50. <https://doi.org/10.1007/s10914-017-9417-6>

2017

Montoya-Bustamante, S., **González, B.**, Zapata-Mesa, N., Obando-Cabrera, L. First records of *Sturnira bakeri* Velazco & Patterson, 2014 (Chiroptera: Phyllostomidae) from Colombia. *Check List*, 13(2), 2091. <https://doi.org/10.15560/13.2.2091>

2016

Martin, G. M., **González, B.** Observations on the behavior of *Caenolestes fuliginosus* (Tomes, 1863) (Marsupialia, Paucituberculata, Caenolestidae) in captivity. *Journal of Mammalogy*, 97(2), 568-575. <https://doi.org/10.1093/jmammal/gyv203>

Technical reports and assesments

IN PRESS

Gaiarsa, M. P., **Gonzalez, B.**, Argueta-Guzmán, M., Salerno, V., Tang, A., Mora, J., Peterson, S. S., McFrederick, Q., Crall, J. Pollination in a changing world: The effects of habitat and temperature on the health of California's wild pollinators (Publication No. CNRA-CCC5-2026-XXX). California's Fifth Climate Change Assessment, California Natural Resources Agency.

2026

Martin, G.M., **González, B.** *Caenolestes caniventer*. The IUCN Red List of Threatened Species 2026: e.T40521A291090573. <https://dx.doi.org/10.2305/IUCN.UK.2026-1.RLTS.T40521A291090573.en>

Martin, G.M., **González, B.** *Caenolestes condorensis*. The IUCN Red List of Threatened Species 2026: e.T136743A291090706. <https://dx.doi.org/10.2305/IUCN.UK.2026-1.RLTS.T136743A291090706.en>

2025

Martin, G.M., **González, B.** *Rhyncholestes raphanurus*. The IUCN Red List of Threatened Species 2025: e.T19710A280099282. <https://dx.doi.org/10.2305/IUCN.UK.2025-2.RLTS.T19710A280099282.en>

Martin, G., **González, B.**, Chemisquy, M. A. *Rhyncholestes raphanurus*. En: Actualización de las categorías de conservación de los mamíferos de Argentina según su riesgo de extinción. Sociedad Argentina para el Estudio de los Mamíferos. Versión digital: <https://cma.sarem.org.ar>

Skills

Programming	R, Python, JavaScript, LaTeX
GIS	R, Python, QGIS, GEE, ArcGis
Software	InkScape, GIMP2, PhotoShop, Illustrator
Back-end	Shiny, Python, Flask, SQL
Front-end	Shiny, React, HTML5
Languages	Spanish, English

Professional Experience

GIS expert

SPRINGER NATURE

Switzerland
May. 2024 - PRESENT

- Creation of figure maps for the project Handbook of the Mammals of Middle and South America” involving 10 volumes and more than 2,000 species.
- Automatization of GIS analysis, creation of range maps, and standardization of map creation within R.

GIS Expert and DevOps

SOCIEDAD ARGENTINA PARA EL ESTUDIO DE LOS MAMÍFEROS (SAREM)

Argentina
Jan. 2024 - PRESENT

- Part of the scientific and organization committee.
- Creation of protocols for standardizing conservation data, including GIS.
- Development of Shiny web app with user-specific permits for editing, revising, and storing biodiversity data.
- Integration of Shiny, R scripts, Google Drive, and AWS databases workflow for integral solutions.

Associate Editor

MAMMALOGY NOTES JOURNAL

Colombia
Mar. 2021 - May. 2025

- Section Editor in charge on managing, reviewing and publishing articles through the Open Journal systems.
- Style editor and reviewer for articles submitted on mammalian biology.

DevOps Programmer

SOCIEDAD ARGENTINA PARA EL ESTUDIO DE LOS MAMÍFEROS (SAREM)

Argentina
Mar. - Nov. 2023

- Report automation for the assessments of Argentinian mammals conservation status.
- Integration of scientific, GIS, and media data for automation report generation.
- Security management for licensed content and distribution of data.

GIS Programmer

SOCIEDAD ARGENTINA PARA EL ESTUDIO DE LOS MAMÍFEROS (SAREM)

Argentina
Jan. - Oct. 2022

- Spatial data cleaning, manipulation, and map design for publication for the threatened mammals of Argentina (CMA-SAREM).
- Generation and integration of software specific solutions to work with data generators and curators.

GIS Programmer and data analyst

SOCIEDAD ARGENTINA PARA EL ESTUDIO DE LOS MAMÍFEROS (SAREM)

Argentina
Aug. 2020 - Feb. 2021

- Spatial data generation, curation, and standardization.
- Development of Shiny web apps for collaborative curation of data.

GIS Programmer and data analyst

FOOD, AGRICULTURE, BIODIVERSITY, LAND-USE AND ENERGY (FABLE) CONSORTIUM

Argentina
Mar. - Dec. 2021

- Analysis of Area of Habitat for all vertebrates of Argentina.
- Integration of Land-use data and biodiversity data for conservation prioritization modeling.
- Shiny app development for generating, curating, and cleaning biodiversity data to calculate Area of Habitat.

GIS Programmer and data analyst

FUNDACION BARILOCHE

Argentina
Sep. 2020

- Biological richness and diversity indices at country level of the threatened terrestrial vertebrates of Belize.

GIS Programmer and data analyst

NATUREMAP AND MINISTERIO DE AMBIENTE Y DESARROLLO SOSTENIBLE

Argentina
Mar. - Dec. 2020

- Range maps refinement to Areas of Habitat for the threatened terrestrial vertebrates of Argentina.
- Documentation, standardization, and creating of metadata.

GIS Programmer and data analyst

MINISTERIO DE AMBIENTE Y DESARROLLO SOSTENIBLE AND SAREM

Argentina
Jan. - Jun. 2020

- GIS database creation and management for the conservation assessment of the mammals of Argentina (CMA-SAREM)

Research consultant

WILDLIFE CONSERVATION SOCIETY

Colombia
Sep. - Oct. 2017

- Promotion and negotiation for the declaration of new private protected areas in the middle valley of the Magdalena river in Colombia for the conservation of the brown spider monkey (*Ateles hybridus*)

Research consultant

FUNDACION AMBIENTE COLOMBIA AND CVC

Colombia
Aug. - Oct. 2017

- Generation of baseline data for the creation of six new private protected areas in La Nevera, Palmira, Valle del Cauca Colombia.
- Trapping mammals with Sherman traps and mist nets.
- Survey local people for community knowledge and identification of key species.

Research consultant

CVC AND PONTIFICIA UNIVERSIDAD JAVERIANA

Colombia
Jul. - Dec. 2017

- Team leader for the control and monitoring of invasive species *Lithobates catesbaianus* and *Lissachatina fulica* in the Valle del Cauca department, Colombia.
- Coordinator for two biologist during the duration of the project.

Research consultant

WILDLIFE CONSERVATION SOCIETY

Colombia
Jan. - May. 2017

- Camera-trap survey and monitoring of *Tapirus terrestris* and other ungulates in the Llanos regions of Colombia.

Research consultant

WILDLIFE CONSERVATION SOCIETY

Colombia
Sep. - Dec. 2016

- Biological characterization of small, medium, and big mammals of the Minas Lili locality in Valle del Cauca.

Teaching and outreach activities

2026	Niches, Species Distributions and Ecosystem Services under Climate Change . Invited talk at Essig Museum. University of California, Berkeley.	USA
2026	Multiple global change drivers reshape the spatiotemporal dynamics of bee-plant interactions: priorities for California's blue orchard bee . Invited talk at MathBio Seminar. University of California, Merced.	USA
2025	Spatial Analyses . Invited talk at INSITE. University of California, Merced.	USA
2025	Best Programming Practices . Open-talk. University of California, Merced.	USA
2025	Species Distribution Modeling . Undergraduate course. Invited Lecturer. University of California, Merced.	USA
2023	Mammalogy . Graduate course. "San Juan Bosco" National University of Patagonia.	Argentina

Grants & Awards

2024	Awarded - not accepted . Postdoctoral fellowship. Three years. \$ 36,000	Argentina
2022	Awardee . Grant for the course "Python programming". \$ 120	Argentina
2019	Fellowship . Collection study grant. Field Museum of Natural History. \$ 3,000	USA
2019	Fellowship . Collection study grant. Smithsonian Institution. \$ 2,900	USA
2018	Awardee . PhD Grant National Council of Sciences and Technology. \$ 43,200	Argentina
2018	Recognition award . V Congreso Colombiano de Zoología. \$ 180	Colombia
2016	Grant . Idea Wild grant for field equipment. \$ 886	USA
2014	Honor . Universidad del Valle. Academic excellence Top 5 best cohort's GPA. \$ 700	Colombia
2012	Honor . Universidad del Valle. Academic excellence Top 5 best cohort's GPA. \$ 700	Colombia

Conference and Talks

2026 — 12th Biennial Conference of The International Biogeography Society

Aarhus, Denmark

- **González, B.**, Guimaraes, P., Gaiarsa, M. From Tropics to Temperate: Latitudinal Vulnerability of Seed-Dispersal Networks under Climate Change. (Poster)

2025 — American Naturalist Asilomar

California, USA

- **González, B.**, Goin, F.J., Martin, G.M. Unraveling the evolution of Paucituberculata: A revised taxonomy and new biogeographical perspectives on Andean marsupials
- Argueta, M., **Gonzalez, B.**, Gaiarsa, M. Projections of heat waves in California and their implications for pollination services.

2025 — 61st Annual Meeting of the Association for Tropical Biology and Conservation

Oaxaca, Mexico

- **González, B.**, Gaiarsa, M. Climate Stability Shapes Seed Dispersal Networks and Functional Persistence in a Biodiversity Hotspot. (Invited talk).

2025 — Orchard Bee Association Meeting

Seattle, USA

- **González, B.**, Gaiarsa, M. Building Climate-Resilient Pollination Systems: Priorities for California's Blue Orchard Bee.

2022 — XXXIV Jornadas Argentinas de Mastozoología

Buenos Aires, Argentina

- **González, B.**, Brook, F., Martin, G.M. 45 años de huella humana en Colombia: impacto sobre los marsupiales Didelphimorphia.
- Martin, G.M., **González, B.**, Brook, F. Distribuciones extremas... ¿Climas extremos? Límites en la distribución de los marsupiales en Argentina. Argentina. Misiones. SAREM. Oral presentation.

2021 — XXXIII Jornadas Argentinas de Mastozoología

Buenos Aires, Argentina

- **González, B.**, Brook, F., Martin, G.M. Marsupiales de Argentina: Idoneidad ambiental y su representatividad en las áreas protegidas del país.
- Martin, G.M., **González, B.**, Brook, F. Riqueza, distribución y conservación de los marsupiales (Didelphimorphia, Microbiotheria, Paucituberculata) que habitan la República Argentina.
- Brook, F., **González, B.**, Martin, Gabriel M., Verzi, D. Patrones morfológicos de los Tucu-Tucus (Rodentia, Ctenomyidae) de Patagonia. Argentina. (Poster)

2019 — XXXII Jornadas Argentinas de Mastozoología

Buenos Aires, Argentina

- **González, B.**, Brook, F., Martin, G.M. Distribución y conservación de las especies de Marmosini (Didelphimorphia, Didelphidae) en Colombia.
- Brook, F., **González, B.**, Verzi, D., Martin, G.M. Una nueva población de *Ctenomys* de los alrededores de Esquel relacionada a *C. sociabilis* Pearson y Christie 1985. (Poster)

2018 — V Congreso Colombiano de Zoología and III Congreso Colombiano de Mastozoología.

Bogotá D.C., Colombia

- **González, B.**, Martin, G.M. Los paucituberculata del norte: modelos de distribución para *Caenolestes fuliginosus* y *Caenolestes convelatus* (Caenolestidae, Paucituberculata).
- **González, B.**, Martin, G.M. ¿Dónde están los marsupiales en Colombia? Un análisis de su extensión de distribución, ocurrencia y solapamiento con áreas protegidas. Colombia.
- **González, B.**, Soria-Escobar, A.M., Pustovrh, M.C., Salazar Monsalve, L., Rojas-Díaz, V. Primera descripción de un embrión del orden Paucituberculata (Marsupialia): El Ratón Runcho (*Caenolestes fuliginosus*).
- Valenzuela-Ospina, L., Herrera-Victoria, A.M., **González, B.**, Saavedra, C.A., Forero-Medina, G. Foto trapeo y patrones de ocupación de vertebrados en el Magdalena Medio y la Orinoquia Colombiana.

2015 — II Congreso Colombiano and III Congreso Latinoamericano de Mastozoología

Bogotá D.C., Colombia

- **González, B.**, Montoya Bustamante, S., Zapata-Mesa, N., Obando-Cabrera, L. Primeros registros de *Sturnira bakeri* (Chiroptera: Phyllostomidae) para Colombia. Colombia
- Ascuntar-Osnas, O., Montoya-Bustamante, S., **González, B.** Ectoparásitos de murciélagos (Diptera: Streblidae) en un fragmento de bosque seco tropical y su distribución potencial En Colombia.

2014 — XXVII Jornadas Argentinas de Mastozoología

Esquel, Argentina

- Martin, G.M., **González, B.** Observaciones sobre el comportamiento en cautiverio de *Caenolestes fuliginosus* (Tomes, 1863) (Paucituberculata, Caenolestidae).

2014 — IV Congreso Colombiano de Zoología

Bogotá D.C., Colombia

- Ascuntar-Osnas, O., Montoya-Bustamante, S., **González, B.** Registros de Streblidae (Diptera: Hippoboscoidea) en un fragmento de bosque seco tropical (Cauca: Colombia).

Complementary Education

2022	Python Programming. Graduate course	Argentina
2021	Introduction to geometric morphometrics in R. Graduate course	Argentina
2021	Species Distributions Modeling with Remote Sensing. Professional course	USA
2021	Taxonomic studies design. Graduate course	Argentina
2021	Biogeography. Graduate course	Argentina
2020	Introduction to geometric morphometrics. Graduate course	Argentina
2019	Origin and evolution of skull and dentition of mammals. Graduate course	Argentina
2019	Brief introduction to the Philosophy of Biology. Graduate course	Argentina
2018	Morphometric and morphogeometric data in phylogenetic studies. Graduate course	Chile
2018	General Linear Models in R. Graduate course	Chile
2016	Camera traps methods for conservation studies on medium and large size mammals. Professional course	Colombia